

The empirical recurrence for OEIS A237892: recovering a conjecture that is not written down

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Abstract

OEIS A237892 records “Empirical recurrence of order 47”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 156$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 47, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 47 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237892 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the maximum minus the upper median of every 2×2 subblock equal.”. It has offset 1 and begins

1848, 32404, 563534, 10390072, 192679586, 3656111168, 69880910510, 1354602459492, ...

The entry states:

Empirical recurrence of order 47 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:32:49, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 156$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 156$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 312$ exact terms of the model returns order 47 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{47} c_i a(n-i),$$

c_1	62	c_{13}	10021246907709	c_{25}	3870239213375413728	c_{37}	−915656559784907929
c_2	−1145	c_{14}	113941299484248	c_{26}	−12963297094512860560	c_{38}	−2189644040967854080
c_3	−4015	c_{15}	−412840967285495	c_{27}	−5664955966438169584	c_{39}	1835674733797232128
c_4	383852	c_{16}	−1008101938004709	c_{28}	32606571649242881176	c_{40}	7040621509842425856
c_5	−3680952	c_{17}	6855228218809292	c_{29}	2480623385975901888	c_{41}	507462521605609472
c_6	−18457082	c_{18}	1765336887356429	c_{30}	−58069250056462812896	c_{42}	−132056748226590720
c_7	503621981	c_{19}	−65453666549209182	c_{31}	7480870294724021696	c_{43}	−333297172917915648
c_8	−1580676045	c_{20}	53041024469186198	c_{32}	74343569053176476352	c_{44}	82634458145882112
c_9	−22629861466	c_{21}	393978040473440396	c_{33}	−17190455248457969728	c_{45}	48168713329459200
c_{10}	171994178529	c_{22}	−613415506563754210	c_{34}	−68838813140007288448	c_{46}	7655936741867520
c_{11}	263617129716	c_{23}	−1541096290052114160	c_{35}	17327891704624158208	c_{47}	419046624460800
c_{12}	−6306561400978	c_{24}	3533599971915524892	c_{36}	46033268683623180160		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 47, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $47 + 4$ terms removed returns order 47 as well, so $\deg q_{\min} = 47 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $47 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 47 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A237892.
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